

TREVOR TEBAWESWA

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EDUCATION

AI App Development Bootcamp

Kampala, Uganda

African Talent Academy (Askar Sulaimanov)

May 2025–September 2025

- Built and deployed AI-powered applications using FastAPI, Docker, Streamlit, LangChain, and LangGraph.
- Designed a food ordering online agent as the capstone project, integrating conversational AI with backend APIs and containerised deployment for real-world functionality.

Makerere University

Kampala, Uganda

BS in Electrical Engineering (National Merit Scholarship)

2022-2025

- CGPA above 4.0 (4.11/5.0)

GLOBAL PROGRAMS & FELLOWSHIPS

Seeds For The Future (Flagship Talent Development Program)

Shenzhen, China

Huawei

July 2025

- Selected for Huawei's global flagship technology leadership program, gaining structured exposure to 5G, AI, cloud computing, and digital infrastructure at enterprise scale.
- Engaged in leadership and entrepreneurship roundtables with senior Huawei executives, gaining insight into technology-driven business decision-making.
- Explored AI agents, hybrid cloud systems, smart cities, renewable energy, and EV technologies, linking system design choices to real-world deployment constraints.

Certificates

Coursera

Kampala, Uganda

John Hopkins University

July 2024 – August 2024

- HTML, CSS and JavaScript for Web Developers
- Introduction to CSS3

University Of Cape Town

Cape Town, South Africa

January 2020 – December 2020

- Completion of CSC1016S (Introduction to Java)
- Completion of CSC1017S (Introduction to Python)

WORK EXPERIENCE

Sunbird AI

Kampala, Uganda

Machine Learning Fellow

July 2025 – June 2026

- Developed an urban noise classification model using transfer learning from YAMNet, achieving 81% accuracy. Deployed the model on Raspberry Pi edge devices across five noise monitoring sensors for real-time classification. The work resulted in a peer-reviewed publication in Environmental Monitoring and Assessment.
- Improved the quality of the Urban-Noise-Dataset-61K by identifying and correcting mislabeled audio samples from Entebbe. Leveraged the noise classification model to generate predicted labels that supported the relabeling and validation of over 24,000 audio recordings.
- Generated and curated synthetic multilingual datasets via machine translation for over 10 low-resource Ugandan languages, producing multiple-choice question-answer pairs and reasoning traces to support the fine-tuning and reinforcement learning of Sunflower.

- Developed an internal leaderboard system for evaluating and tracking model performance across diverse metrics and benchmarks such as AfroBench, supporting comparison between internal models and external open- and closed-source systems.
- Built a multilingual RAG-based citizen feedback analytics platform for public service institutions, using Sunflower's transcription and natural language understanding capabilities to analyse sentiment and feedback from hospitals, police stations, and municipalities. The solution was deployed and handed over to SEMA for use in public service monitoring and accountability.
- Fine-tuned medical-domain LLM adapters for the HASH project, achieving over 15% improvements on semantic and non-semantic evaluation benchmarks. Implemented hot-swappable adapter deployment in an offline-first application delivering sexual and reproductive health information to youth in five African countries.
- Conducted reinforcement learning research using Group Relative Policy Optimisation (GRPO) on Sunflower, Uganda's first multilingual large language model, achieving over 15% gains in factuality, 10% improvements in instruction following and translation quality, and measurable reductions in model glitching and unstable outputs.

Kiira Motors Corporation

Research Associate I

Jinja, Uganda

March 2025 – July 2025

- Built a charging scheduling algorithm based on a predicted energy consumption model for the Jinja- Iganga under the E-bus Xpress project to minimise the costs for battery degradation by 33%.
- Developed an energy consumption model using supervised machine learning with an accuracy of 91.73%.

Kiira Motors Corporation

Undergraduate Intern

Nakasongola, Uganda

May 2024 – July 2024

- Contributed to the production of twenty-eight buses through the installation and testing of front and rear powertrain and electrical engineering system components for 7 units of 10.5-meter EV buses, three units of 6.6-meter EV buses, and 3 units of 8.5-meter EV buses.
- This led to thirteen dynamic tests being carried out, during which the buses covered a combined distance of 700 km without any failure of powertrain or electrical engineering systems.
- Demonstrated strong technical aptitude when troubleshooting problems related to the front and rear powertrain and electrical engineering system components of the first 6.6-meter EV bus, which led to a 40% increase in the production rate of the 6.6-meter EV buses.

Sunbird AI

ML Engineer Intern

Kampala, Uganda

July 2023 – September 2023

- Developed a language identification model using supervised machine learning up to a percentage accuracy of 99.93%, which was implemented into the current machine translation systems.
- Cleaned and uploaded text-to-speech audio data of 5000+ sentences in Acholi and Luganda to Sunbird AI's open-source data servers.

VOLUNTEERING

Creative Child Foundation

Volunteer

Kampala, Uganda

November 2024 – January 2025

- Facilitated over four hours at two secondary schools and one primary school, familiarising over 150 students with OctoStudio, a mobile-based creative coding platform designed for children, and introducing students to coding fundamentals. The children were guided to create projects on the SDGs, such as environmental conservation, quality education, and good health and well-being.

PERSONAL PROJECTS

Pinecone Venture Studio

Kampala, Uganda

Full-stack Developer & ML Engineer

December 2025 – March 2026

- Built an AI-powered venture validation platform for early-stage startup evaluation, combining a multi-agent LLM pipeline with a custom-trained XGBoost classifier to predict venture success likelihood from historical startup outcome data.
- Designed the end-to-end data preprocessing and feature engineering workflow, merging multi-source datasets, deriving funding-based features, and constructing outcome labels via an exit-multiple valuation model and then packaged the trained pipeline for production inference.
- Implemented the full-stack application using React, TypeScript, and Firebase, integrating LLM providers and live web research APIs to ground agent outputs in real-time data.

Integrated Smart Systems Group

Kampala, Uganda

Undergraduate Researcher

August 2023 – March 2024

- Developed a time series model made up of a combination of two time series models (SARIMAX and Holt-Winters' Exponential Smoothing) using supervised machine learning, with a percentage accuracy of 99.955% to forecast long-term electricity demand in Uganda up to 2040.

LEADERSHIP EXPERIENCE

Makerere University

Kampala, Uganda

Class Representative

2022–2025

- Collaborated with other class representatives at the college to replace blackboards with whiteboards in over 15+ classrooms.

SKILLS & INTERESTS

Technical Skills

- *Programming Languages:* Python, C, C#, C++, R, Java, Javascript
- Containerization: Docker
- Microsoft Office Suite (Word, Excel, PowerPoint, Access), Google Suite

Proficient Languages

- English, Luganda

